KARNAVATI UNIVERSITY	**Karnavati University** **Unitedworld Institute of Technology** **Gandhinagar, Gujarat - 382422**	**UIT**

Course Syllabus

School	**Unitedworld Institute of Technology**	**Programme Code**	**1039**
Programme	**B.Tech.**	**Branch**	**CSE, CSE-AIML**

Sem VII: - Year of Admissions/Batch: 2023-2027

Course code	**Course Name**	**Hours/week**			**Credit**	**Total marks**
		L	**T**	**P**	**C**	
71407004E01	**Agentic AI**	2	0	2	4	**100**

Course Outcomes	**Statements**	**Bloom's Level**
CO1	Recall and explain what AI Agents are — autonomy, perception, memory, and action; distinguish agents from chatbots.	L2
CO2	Understand the ReAct framework, tool-use patterns, and memory systems used in agentic AI.	L2
CO3	Apply LangChain / LlamaIndex to implement RAG pipelines and single-agent workflows.	L3
CO4	Analyse multi-agent architectures, evaluate agent performance, and apply guardrails for safe operation.	L4
CO5	Design and deploy an end-to-end agentic AI application integrating LLMs, tools, memory, and RAG.	L6

Prerequisite	Python Programming, Data Structures, Basic Mathematics (Linear Algebra, Probability)		
Evaluation Scheme	**Theory + Practical**	**Hours**	**Marks**
	End Semester Examination	2 hours	50
	Internal (Theory) – 50 Marks ● Mid-Sem Exam. – 25 Marks ● CIA (Certification/Quiz/Assignment) - 20 Marks ● Attendance - 5 Marks **Internal (Practical) – 50 Marks** ● Mid-Sem Exam. – 25 Marks ● CIA (Labs/Mini Project) - 20 Marks ● Attendance - 5 Marks **Note:** L+ P Bifurcation will be credit-based out of 50		50

Course Description	This course introduces students to the fundamentals of Agentic AI systems. Students learn how LLM-powered agents reason, use tools, maintain memory, and collaborate in multi-agent workflows. Emphasis is on hands-on implementation using LangChain, OpenAI API, and vector databases, culminating in a deployable end-to-end agentic application.	
UNIT-I	**FOUNDATIONS OF AGENTIC AI**	**9**
What is an AI Agent — autonomy, perception, memory, action; agents vs. chatbots vs. assistants. ReAct Framework — Reasoning + Acting loop; thought-action-observation cycle with examples. Tool Use — function calling (OpenAI / Gemini), tool schemas, and structured JSON outputs. Memory Systems — in-context memory, external vector memory, episodic vs. semantic memory. Planning in Agents — task decomposition, sub-goal generation, Plan-and-Execute pattern.		
UNIT-II	**LLM FRAMEWORKS & RAG**	**9**
LangChain / LlamaIndex — chains, agents, tools, document loaders, text splitters, retrievers. Vector Databases — FAISS, ChromaDB, Pinecone — indexing, similarity search, metadata filtering. Retrieval-Augmented Generation (RAG) — ingestion pipeline, chunking strategies, retrieval-augmented prompting. Embeddings in Agents — OpenAI / HuggingFace embeddings, cosine similarity, semantic retrieval. Agent Memory with Vector Stores — persisting and querying agent memory across sessions.		
UNIT-III	**MULTI-AGENT SYSTEMS & ADVANCED PATTERNS**	**9**
Multi-Agent Systems — orchestrator-worker pattern, agent roles, communication protocols. AutoGen / CrewAI — multi-agent frameworks, defining agents, tasks, and flows. Agentic Patterns — reflection, self-critique, tool chaining, parallel agent execution. Code Generation Agents — using LLMs for code writing, debugging, and code review automation. Human-in-the-Loop — approval workflows, human feedback integration, escalation patterns.		
UNIT-IV	**AGENT EVALUATION, SAFETY & GUARDRAILS**	**9**
Agent Evaluation — trajectory evaluation, tool call accuracy, task completion rate. Guardrails — output validation, content moderation, safety layers (NeMo Guardrails overview). Observability — LangSmith / LangFuse for tracing agent runs, cost tracking, latency monitoring. Failure Modes — hallucination in agents, infinite loops, tool misuse; mitigation strategies. AI Ethics & Governance — GDPR implications, data privacy in LLM apps, model cards, transparency.		
UNIT-V	**DEPLOYMENT & INTEGRATION**	**9**
Deploying AI Applications — FastAPI backend + React frontend, REST endpoints for LLM pipelines. Containerisation — Docker basics for AI app packaging, environment management. Cloud Deployment — Hugging Face Spaces, Render, Railway for hosting AI apps. Capstone Project Design — scoping, system design, tech stack selection, demo presentation. End-to-End Integration — connecting LLM, tools, memory, RAG, and deployment in one pipeline.		
Total hours	**45 hours**	

CO- PO/PSO Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO 1	-	2	3	-	3	1	-	-	2	1	-	2
CO 2	-	2	3	-	3	1	-	-	2	1	-	2
CO 3	-	2	3	2	3	2	-	-	2	1	-	2
CO 4	-	2	3	2	3	2	-	-	2	1	-	2
CO 5	-	2	3	2	3	2	-	-	2	2	2	2

List of Practicals:

Sr. No.	Practical	CO	Hours
1	Install Python, VS Code/Jupyter, API Keys, and build your **first AI agent** that answers simple user queries. Understand the difference between a chatbot and an AI agent.	CO1	2
2	Create a simple **ReAct Agent** capable of reasoning and using one basic tool (calculator or current date/time).	CO1	2
3	Implement **conversation memory** so the agent remembers previous interactions and provides contextual responses.	CO2	2
4	Create a small **Knowledge Base** using PDF/Text files and build a basic **RAG application** using LangChain and FAISS/ChromaDB.	CO2	4
5	Develop a **Document Question Answering Agent** using the created knowledge base.	CO3	2
6	Build a simple **Multi-Agent Workflow** consisting of a Planner Agent and an Executor Agent working together.	CO3	4
7	Implement **Guardrails** by validating outputs, restricting unsafe prompts, and handling invalid responses.	CO4	2
8	Evaluate the developed agents using task completion, response quality, latency, and accuracy metrics.	CO4	2
9	Deploy the developed AI agent using **FastAPI** and test it through REST APIs or a simple web interface.	CO5	4
10	**Mini Project:** Design and deploy a basic AI Agent for any one domain (Student Assistant, Library Assistant, College FAQ, Banking FAQ, Healthcare FAQ, Shopping Assistant, etc.) using tools, memory, and/or RAG. Demonstrate and evaluate the complete solution.	CO5	6
	Total	**30 hours**	

TEXT BOOKS:

1	Ben Auffarth — Generative AI with LangChain, Packt Publishing, 2023
2	James Phoenix & Mike Taylor — LangChain Crash Course: Build OpenAI LLM-powered Apps, Independently Published, 2023
3	Maxime Labonne — LLM Engineer's Handbook, Packt Publishing, 2024
4	Denis Rothman — Transformers for Natural Language Processing, 2nd Ed., Packt Publishing, 2022
5	Andrew Ng — AI for Everyone (Course Notes + Supplementary Text), DeepLearning.AI, 2023

REFERENCES:

1	Chip Huyen — Designing Machine Learning Systems, O'Reilly Media, 2022

2	Valentina Alto — Modern Generative AI with ChatGPT and OpenAI Models, Packt Publishing, 2023
3	Omar Sanseviero et al. — The Hugging Face NLP Course (online + print edition), HuggingFace, 2023
4	Michael Walmsley — Agentic AI: Patterns, Architectures and Best Practices, independently published, 2024
5	Simon Ellin — Building AI Agents with Python: Practical Guide to LangChain, Tools & Memory, Independently Published, 2024
LINKS AND ADDITIONAL READING MATERIAL:	
1	OpenAI API Documentation https://platform.openai.com/docs
2	LangChain Python Documentation https://python.langchain.com
3	HuggingFace Model Hub & Docs https://huggingface.co/docs
4	DeepLearning.AI Short Courses (LangChain, RAG, Agents) https://www.deeplearning.ai/short-courses/
5	Anthropic Claude API Documentation https://docs.anthropic.com